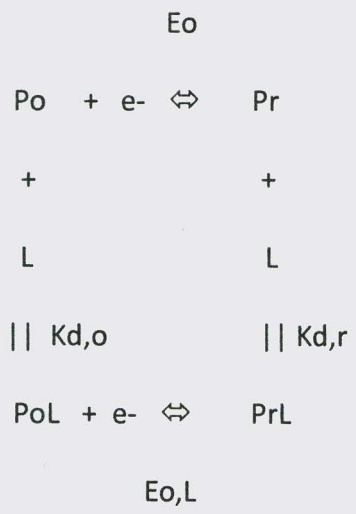
Biochemistry Cume
March 29, 2013

Consider a redox protein, P, for which both the oxidized (Po) and reduced (Pr) states can bind a ligand L. The dissociation constant for binding of L to the oxidized state is Kd,o and for binding to the reduced state is Kd,r.



Derive an equation relating the difference between the two redox potentials (Eo,L - Eo) to the ratio of the dissociation constants (Kd,o/Kd,r)

What will be the general relationship (which will be greater) between the redox potentials when

Kd,o > Kd,r

Kdo < Kd,r

Kd,o = Kd,r